



Utthan Singh Roy

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Online BSc (Hons.) - Data Science and AI

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EDUCATION

Degree Discipline	Institute/Board	CGPA/Percentage	Year
BSc (Hons.) Data Science and AI	Indian Institute of Technology, Guwahati	8.24 (Current)	2023-Present
B.Tech Electrical Engineering	Madan Mohan Malaviya University of Technology, Gorakhpur	8.36 (Current)	2023-Present
Senior Secondary	CBSE Board, St. Joseph's School, Shaktinagar	94.5%	2021
Secondary	CBSE Board, St. Joseph's School, Shaktinagar	93.8%	2019

EXPERIENCE

- Indian Institute of Technology (IIT) Guwahati** *Jul. 2026 – Present*
LLM Application Engineering Intern *Guwahati, Assam*
 - Architected a low-latency Hybrid RAG pipeline combining ChromaDB/BM25 retrieval with asynchronous LCEL token streaming to minimize Time-to-First-Token (TTFT).
 - Engineered citation and deterministic fallback guardrails for LLM outputs, improving reliability of production-oriented microservice responses.

PROJECTS

- AI-Powered Market Regime Shift Classifier** *Feb. 2026 – Jun. 2026*
Python, XGBoost, FinBERT, Streamlit, GitHub Actions *GitHub*
 - Engineered a cost-sensitive XGBoost model on 41 quantitative and FinBERT sentiment features, achieving 0.97 ROC-AUC for systemic market stress detection.
 - Built a real-time inference dashboard and automated feature engineering, deployment, and model updates through GitHub Actions CI/CD.
- Physics-Validated AI Architecture for Autonomous Grid Resilience** *Apr. 2026 – Jul. 2026*
Python, LightGBM, SAC Reinforcement Learning, LangGraph *GitHub*
 - Designed Soft Actor-Critic (SAC) reinforcement learning policies with a LightGBM forecasting layer achieving 1.43% MAPE and 100% convergence in AC power-flow simulations.
 - Implemented a deterministic validation firewall for LLM-generated targets, dynamically rejecting invalid outputs and improving the Grid Resilience Index by 6.41 points.
- Automated Sustainable Portfolio Optimizer** *Oct. 2025 – Jan. 2026*
Python, XGBoost, NSGA-II, GitHub Actions, Pandas *GitHub*
 - Combined NSGA-II optimization with XGBoost to forecast 30-day Nifty 50 directionality and dynamically optimize asset weights under tail-risk constraints (CVaR).
 - Automated daily data ingestion, model retraining, and deployment through a GitHub Actions pipeline.

TECHNICAL SKILLS

- Programming:** Python, SQL, C, Java, R
- AI/ML:** PyTorch, TensorFlow, Scikit-Learn, XGBoost, LightGBM, SAC, NSGA-II, FinBERT, SHAP
- LLM/Agentic AI:** LangGraph, RAG, LCEL, ChromaDB, BM25, API Integration
- MLOps/Tools:** Git, GitHub Actions, Streamlit, SQLite, Pandas, NumPy

KEY COURSES TAKEN

- Computer Science & AI:** Data Structures, Algorithm Design & Analysis, Deep Learning for Computer Vision, Hardware-Aware Deep Learning, LLM Application Engineering
- Mathematics & Analytics:** Linear Algebra, Optimization, Statistical Inferencing, Time Series Analysis
- Systems & Engineering:** Control System Engineering, Digital Control System, Microprocessors

POSITIONS OF RESPONSIBILITY

- Volunteer,** AI Confluence, IIT Guwahati *2024*

CERTIFICATIONS & ACHIEVEMENTS

- Explainable AI (XAI) Specialization,** Coursera, Duke University *2026*
- AI in Healthcare Specialization,** Coursera, Stanford University *2026*
- Qualified GATE 2026 – Data Science & Artificial Intelligence (DA),** 2026